

4 Introduction to programming

A program or a code is a predefined set of instructions which are fed into the memory of a processing unit. These instructions define what the processing unit would do under different input conditions. This set of instructions is called a 'code' or a 'program'.

If you have to instruct someone to do some task, you would need to have a medium through which you can convey your idea. The medium can be a language or a sign or something else, but it must be understood by you as well as the person whom you want to instruct. Similarly, to instruct a microcontroller you need a common language or a platform, where you can exchange instructions or information.

Generally, all the machines understand 'binary language', which is composed of only 0's and 1's. Unfortunately, it is very difficult for us to convey instructions in binary form for us. And machines also do not understand any of the languages we speak.

A common language, thus, is required. These common languages are called 'programming languages' and the most commonly used are C, C++ and JAVA. Here we would use C++.

Given below is a sample C code for sampling data over RS-232 interface.

```
////////////////////////////////////  
/// This program displays the min and max of 30,    ///  
/// comments that explains what the program does,  ///  
/// and A/D samples over the RS-232 interface.      ///  
////////////////////////////////////  
  
#if defined(__PCM__) // preprocessor directive that  
chooses the compiler  
#include <16F877.h> // preprocessor directive that  
selects the chip PIC16F877  
#fuses HS,NOWDT,NOPROTECT,NOLVP // preprocessor directive  
that defines fuses for the chip  
#use delay(clock=20000000) // preprocessor directive  
that specifies the clock speed  
#use rs232(baud=9600, xmit=PIN_C6, rcv=PIN_C7) //  
preprocessor directive that includes the rs232 libraries
```